

Reliability Framework Integrating Grid Scale BESS Considering BESS Degradation

Mohamed Galeela^{a,*}, Konstantinos Kopsidas^b, Diptargha Chakravorty^c

^a Department of Electrical Engineering, Faculty of Engineering, Cairo University, Cairo, Egypt

^b Department of Electrical and Electronic Engineering, Engineering School, University of Manchester, UK

^c TNEI Services LTD, Manchester, UK

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ABSTRACT

To ensure the future networks' secure and flexible operation, NOs will increasingly rely upon grid-scale storage services as a secondary (ancillary) service. During critical contingencies, NOs may drive some battery energy storage systems (BESS) that operate at higher depth-of-discharge cycles for increasing reliability benefits. This eventually could lead to accelerated degradation of BESS. The reliability benefits versus the accelerated BESS degradation risks over the BESS lifetime are not deeply investigated in the up-to-date literature. This paper develops a reliability evaluation framework integrating a Bi-Level Sequential Monte Carlo loops with a detailed BESS degradation model. The integration captures the BESS normal and accelerated cycle and capacity degradation across a multi-year network analysis. Annualized indices are defined for assessing the risks of BESS accelerated degradation named as expected equivalent life cycle accelerated degradation (EELCAD), expected equivalent capacity degradation (EECD) and expected battery accelerated degradation costs (EBADC). In addition, it provides a techno-economic assessment of reliability benefits versus the degradation risks over the battery's lifetime. Therefore, the paper presents an in-depth analysis of the benefits and risks associated with battery degradation and the potential implications in long-term planning studies.

1. Introduction

Power networks are currently committed to delivering a high quality of supply in an economic way preserving high security and reliability levels considering network uncertainties. They have to fulfill the above commitments while tackling the current challenges of continuing demand growth, change in demand characteristics and high penetration of renewable energy resources (RES) which add more stresses on network components affecting the network reliability levels. To overcome such challenges, network reinforcement and expansion are required; however, many economic and environmental constraints restrict such actions. This necessitates network operators (NOs) to explore alternate approaches that provide flexible methods for increasing the network capacity while deferring network reinforcement. Energy storage is currently utilized by NOs to improve network reliability and face the above challenges.

The coordination between wind and Energy Storage System (ESS) showing the impact on system's adequacy is investigated in [1]. The authors in [2] investigated the impact of variations in energy storage

capacity on the system's reliability indices. The impact of energy storage on minimizing the wind curtailments and system's energy not supplied (ENS) is investigated in [3]. The impact of different operating parameters of energy storage on ENS and expected surplus wind energy is investigated in [4]. These operating parameters are the charging/discharging constraints and energy storage capacity.

Most of the above literature uses either general ESS models without referring to a specific technology or using hydropower, compressed air with the lower amount of BESS. However, the BESS started to have high potential on the grid-scale applications. The authors in [5] present an overview of different BESS technologies showing their potential in power system applications. The utilization of BESS for transmission congestion relief is illustrated in [6]. Also the authors in [7] showed that with BESS, the need for network reinforcement could be postponed. In the context of reliability, the authors in [8] developed an adequacy framework for power networks considering wind farms and BESS showing the role of BESS in minimizing wind curtailments. In addition, a method of optimal placement of BESS for maximizing the network reliability benefits is proposed in [9]. An intelligent operation strategy is presented in [10] showing the impact of utilizing the BESS in

* Corresponding author.

E-mail addresses: Mohamed.galeela@eng.cu.edu.eg (M. Galeela), K.Kopsidas@manchester.ac.uk (K. Kopsidas), diptargha@tneigroup.com (D. Chakravorty).

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Nomenclature

A. Indices

j	Bus number running from 1 to N_b
i	Index represents the percentage of the depth of discharge
y	SMC Simulation year running from 1 to Y

B. Parameters

Cap_{deg}	BESS capacity degradation
DOD_j	Depth of discharge of BESS installed at bus j
Δt	Time step of the annual SMC loop
$P_{j,BESSdis}(t)$	BESS power discharge at time step (t)
$P_{j,BESSch}(t)$	BESS power charge at time step (t)
$\eta_{j,BESSch}$	BESS charge efficiency
$\eta_{j,BESSdis}$	BESS discharge efficiency
$E_{j,BESS}(t)$	Energy stored in the BESS at time step (t)
$Ah_{i,throu}$	Total energy throughput of the BESS
Ah_{cap}	BESS Ampere-hour capacity
$p_{curtailed}(t)$	Power curtailment at bus j
$DOD_{nominal}$	Nominal BESS depth of discharge of the BESS at normal conditions
$DOD_{j,i}$	DOD of percentage i for the BESS installed a bus j
$C_{DODj,i}$	Counter recording number of discharge cycles occurred at this $DOD_{j,i}$
DOD_{base}	Manufacturer depth of discharge
$L_{DODbase}$	Number of life cycles the BESS end of Life (EOL) is expected at DOD_{base}

L_i	Number of life cycles the EOL is expected when BESS is operated at i th DOD_i .
$Cap_{j,deg,acc}$	Accelerated BESS capacity degradation when BESS operated at a DOD higher then the $DOD_{nominal}$
$Cap_{j,deg,norm}$	Normal BESS capacity degradation when BESS operated at a DOD less than or equal then the $DOD_{nominal}$
$C_{j,eq}$	Cumulative equivalent cycle degradation considering different DOD percentages of operation referred to the based DOD.
$P_{j,cap}$	Power rating for the BESS at bus j
$E_{j,cap}$	Energy capacity for the BESS at bus j
SOC	BESS state of charge
$\sigma_{min}, \sigma_{max}$	Minimum and maximum DOD limits as a percentage of the BESS energy capacity.
v_j	Voltage magnitude at bus j
$\mathfrak{R}_{j,Y_i}$	A coefficient capturing sequentially the accelerated degradation history of the BESS at bus j
v_j	Voltage magnitude at bus j
EEACD	Expected equivalent accelerated capacity degradation
EENCD	Expected equivalent normal capacity degradation
EELCAD	Expected equivalent life cycle accelerated degradation
EELCND	Expected equivalent life cycle normal degradation
EBADC	Expected BESS accelerated degradation costs
Y_{Life}	BESS estimated life in years
Y	Number of sequential Montecarlo simulation years

distribution networks to provide reliability services for the transmission network when they are in the grid-connected mode.

One of the main constraints of BESS implementation on a grid-scale level is the high capital cost requirements. Hence, much literature is focused on combining other flexibility methods with BESS to enhance network reliability. Dynamic thermal rating of OHLs (DTR) is considered an efficient solution for increasing the grid's flexibility. The line ratings could be increased by 10–30%, and even up to 50% in windy areas [11]. The reliability performance of the TVTR system and its influence on the reliability performance of transmission systems are estimated in [12]. The authors in [13] examined the impact of dynamic thermal rating with BESS on the reliability of wind-integrated networks by defining new indices to measure the wind enhanced capability of BESS. The same authors expanded their work to consider how DTR and BESS could affect network topology optimization for improved system reliability and maximizing wind power utilization [14]. The previous work focused on how both DTR and BESS enhance network reliability; however, it neglected the risk of ICT failures on DTR monitoring system's unavailability [15,16]. Hence, network reliability is adversely affected. The above literature also neglected the benefits of combining DTR and BESS on reducing the capital costs required from BESS alone to preserve a specific network reliability level.

From BESS lifetime modelling perspective, much work is done in predicting the lifetime of batteries. The authors in [17] used advanced machine learning techniques and experimental setups to have an accurate battery life-time prediction model for Zn-ion batteries with experimental validation. The same authors identified the gaps in the traditional machine learning methods for long-term battery capacity prediction and proposed a new method based on gaussian process regression that proves large accuracy and minimum error in predicting the capacity of zinc-ion batteries [18]. Other authors explored different techniques to extend the life of next generation lithium-sulfur batteries [19]. The above literature comprehensively explored different lifetime prediction models however, these models do not allow to capture the degradation impact on batteries that may go to a deep cycle operation

under network contingencies. Hence, these models' complexity makes it unsuitable to be integrated with network-wide reliability analysis.

In terms of BESS degradation, it is mainly investigated in electric vehicle applications [20,21]. However, there is very limited literature on BESS degradation in the grid-scale level. The impact of BESS degradation on increasing the energy arbitrage revenue of grid-level storage is investigated in [22]. The authors in [23] developed an ageing awareness control strategy for lithium-titanate BESS used in leveling applications. Also, the authors in [24] provide a sizing technique for second life estimation for LiFePO₄ for facilitating high wind integration to the grid. Considering the BESS operation and planning analysis, some studies assume that the battery has a fixed life period with negligible O&M costs [25,26]. The previous studies shared the same assumption that the battery could only perform one complete cycle per day. These assumptions are not perfectly fitting for the grid-scale power system operation where on some occasions it is required to cycle the BESS more than once daily with different depth of discharges (DOD) at either normal or contingency conditions in the network. However, this leads to the degradation of the battery reducing its lifetime.

To address the above problem the researchers in [27] constrained the maximum number of daily cycles the battery could perform. However, this is not considering the full flexibility of the battery. The authors in [28] utilize the flexibility of cycling the battery many times through the day to provide ancillary services and participate in the electricity markets. They consider the battery degradation model due to the variation of DOD per cycle with respect to the electricity market prices. Hence, the battery degradation costs are considered as operating costs in the economic model to define the actual profitability from battery utilization.

So far, none of the above literature integrates the BESS accelerated degradation within the reliability framework. Hence, the lifetime reduction risks of the BESS versus the gained network reliability benefits are investigated. To that extent, this work develops a reliability evaluation framework considering BESS accelerated degradation with the below contributions.

- 1) Development of a generic BESS degradation model capturing the capacity and cycle degradation of the BESS.
- 2) Integration of the developed BESS degradation model within a Markov chain to facilitate a standard Sequential Monte Carlo reliability framework to assess the risks and benefits of BESS degradation.
- 3) Development of three new indices to capture BESS degradation risks named: a) Expected equivalent lifecycle accelerated degradation (EELCAD), b) expected equivalent capacity degradation (EECD) and c) expected battery accelerated degradation costs (EBADC).
- 4) Implementation of a sequential multi-year network reliability analysis considering annual BESS degradation. Hence, the long-term BESS degradation risks versus reliability benefits are quantified over BESS expected lifetime.
- 5) Investigation of BESS accelerated degradation impact on Net Present Value (NPV) profits in an operators' network planning investment analysis.
- 6) Examine dynamic thermal rating partial displacement of BESS size requirement to maintain desired reliability levels.

2. BESS lifetime estimation models

Lifetime estimation for BESS is modeled with two different approaches, one uses performance-based model and the second uses the cycle counting or Weighted ampere-hour (AH) throughput models [29].

2.1. Performance-based models

These models consider the change in the performance parameters of the battery from capacity, voltage, power ...etc. The end of life (EOL) of the battery is reached when those performance parameters drop below a threshold value. They are classified into electrochemical models, equivalent circuit models and heuristic models which require too much input parameters to evaluate the degradation. For the electrochemical models, the input data are related to the chemical reactions of the battery. For the equivalent circuit, the battery is modelled with an equivalent circuit from voltage, current sources, resistances, capacitances, and inductances. The degradation is captured by the variations in the circuit parameters. However; for the heuristic models, the simplicity is that only the historical and current measurements are required without understanding the relationship between input and output data [20].

2.2. Cycle counting and (Ah/Wh) throughput lifetime models

The cycle counting lifetime models are based on capturing the remaining useful life of the battery by summing the incremental lifetime cycles exhibited by the battery. The most dominant factor in cycle counting model is the DOD. Such that as the DOD increases the lifetime is reduced in terms of reducing the number of battery life cycles until its EOL.

The authors in [30] examined an improved rain flow cycle counting algorithm combined with the autoregressive integrated moving average prediction model to propose a new prediction for the Li-ion battery state of health. Also, the authors in [31] proposed an improved rain flow algorithm combined with a linear criterion for accurate Li-ion battery residual life prediction. The authors in [32] proposed a fast battery cycle counting method for grid-connected Battery Energy Storage System (BESS). Hence, different cycle counting had been introduced in the literature for battery lifetime estimation.

Fig. 1 represents the relationship between the number of cycles and DOD for different BESS technologies which are Lead acid, lithium ion, Flow battery, sodium sulphur and Ni-MH [33]. From the figure it is obvious that as DOD increases the number of life cycles till EOL decreases in a non-linear way. The values in Fig. 1 are the mean values of a distributed data after testing a certain number of battery samples. It is also clear that the relationship is common for all technologies which is a

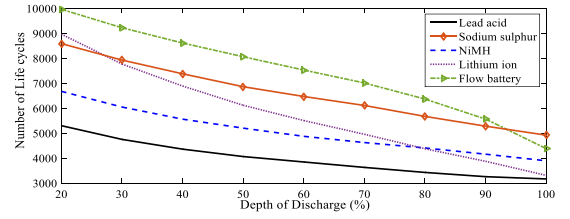


Fig. 1. Life Cycles Vs. Depth of discharge for different BESS technologies.

polynomial decay but with different values depending on the technology type.

On the other hand, the Ampere-hour/watt-hour (Ah/Wh) throughput lifetime models are based on counting the total incremented Ah/Wh energy throughput from the battery at different DOD levels. This captured energy throughput is being compared to the total Ah/Wh throughput, the battery can achieve until its EOL, under certain standard operating conditions such as current rate, DOD and temperature [34]. These standard operating conditions are usually defined by the manufacturers. The battery fails, i.e., reaches its EOL, once the value of total Ah throughput at the standard operating condition is exceeded.

Due to the complexity of the first approach (performance-based models require too many input data), this work considers only the cycle counting degradation modeling. The cycle counting degradation models are used to capture the capacity degradation of the BESS using the manufacturer data that relates the capacity degradation with the number of cycles at a certain DOD. Fig. 2 plots the relationship between the capacity in percentage and the number of cycles at 100 % DOD for a lithium-ion BESS [35].

Fig. 2 could be mathematically represented by a polynomial function with different parameters depending on the BESS technology. This polynomial function is represented by (1).

$$Cap (\%) = (a(C_{eq})^n + b(C_{eq})^{n-1} + c(C_{eq})^{n-2} + \dots + Z) \quad (1)$$

Such that Cap is the capacity in percentage, $N_{DOD-base}$ represents the number of cycles at the base DOD at which Fig. 2 is generated, these number of cycles are obtained from the relationship described in Fig. 1. The variables $a, b, c \dots Z$ depend on the BESS technology and these parameters could be easily captured from the provided manufacturer data. The capacity degradation Cap_{deg} is simply captured by (2).

$$Cap_{deg} (\%) = 100 - (a(C_{eq})^n + b(C_{eq})^{n-1} + c(C_{eq})^{n-2} \dots + Z) \quad (2)$$

The full steps for evaluating the capacity degradation are illustrated in Section 4.

3. Methodology overview

The computational procedure of the proposed reliability evaluation methodology which considers BESS degradation is summarized in Fig. 3. To capture the risks associated with the degradation of BESS for use in emergency operation when an elevated DOD (i.e., higher than the

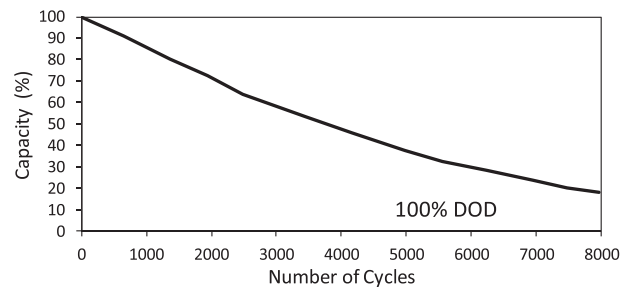


Fig. 2. Capacity % vs. Number of cycles.

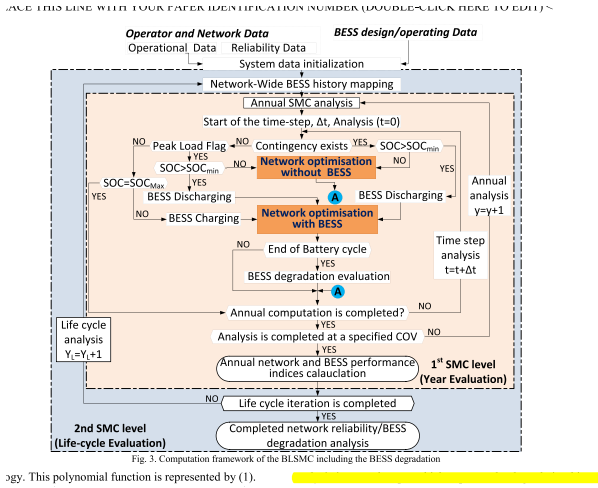


Fig. 3. Computation framework of the BLSMC including the BESS degradation.

nominal values) is permitted, a detailed BESS degradation model is integrated within the network reliability framework. This integration is achieved through a Bi-level sequential Monte-Carlo loop (BLSMC) that can quantify the long-term BESS degradation effect and hence benefits and risks associated with its lifetime reduction. In Fig. 3, the first SMC level calculates the annual network reliability analysis, and the second level calculates the BESS life cycle analysis in annual steps, which captures the degradation history in a sequential manner. In this type of long-term sequential analysis an annual load growth is considered based on the network expected trends.

The calculations are initiated by acquiring the main input data: the network and operator data, the BESS design and operation data. The network data include the reliability (Mean time to fail, MTTF, and mean time to repair MTTR) data and network topology; while the operation data include the bus voltage limits, the generators power characteristics, OHLs ratings and impedances, and the annual chronological hourly demand data. The BESS design and operating data block include the battery technology, the location, the energy capacity, the power rating, and efficiency, as well as the BESS manufacturer nominal operating conditions data from temperature and cycle depth.

The first annual SMC (in Fig. 3) is performed to sample the MTTF and MTTR for the system components and generate their operation and restoration transitions for each time step (Δt) over a simulation year (y).

When no contingency exists, BESS is discharged or charged based on its state of charge (SOC) level and the minimum SOC (SOC_{min}) for providing normal operation services to optimize peak demand. When SOC is below SOC_{min} , then the BESS cannot be utilized within the ACOPT network optimization (as showing in Fig. 3). When peak load does not exist, and SOC is not at its maximum state then BESS is charged as it is dictated by the ACOPT network optimization [36].

When contingency exists, the SOC is updated and if it exceeds the minimum value (SOC_{min}) then an ACOPT network optimization is implemented with the inclusion of BESS, at Δt . The minimization of network production and interruption costs is performed considering network and BESS constraints. Due to contingencies, an elevated DOD level of BESS is considered as an additional corrective action following generation re-dispatch with load curtailments being considered as a last flexibility option. However, when the SOC is below SOC_{min} then optimization, at Δt , neglects the BESS.

Since elevated DOD is only allowed at contingencies, charging of BESS is permitted after the contingency only when the Peak load flag is not exceeded (Fig. 3).

In Fig. 3 the end of the BESS cycle is determined in order to calculate the normal and accelerated degradation parameters and the updated SOC at Δt . The detailed degradation modeling is shown in Section 4.

After the end of the annual SMC level, when the convergence criterion is met, the annual outputs are evaluated. These outputs are (a) the network reliability indices: Expected energy not supplied (EENS), expected interruption costs (EIC) and expected total operating costs (ETOC), (b) the expected annual BESS performance indices: Expected equivalent capacity degradation (EECD), expected equivalent lifecycle accelerated degradation (EELCAD), and expected battery accelerated degradation costs (EBADC). The mathematical formulation of these indices is illustrated in detail in section IV.

The life-cycle SMC loop is implemented for each year (Y_L) of the BESS's design life. This life-cycle loop evaluates chronologically the cumulative values of the annual SMC loop indices for the complete BESS life-cycle analysis (Y_{LC}). Hence, it generates the degradation history mapping for all BESS installed in the network. This loop is terminated once the final year of the life-cycle analysis (i.e., Y_{LC}) is simulated.

4. Bi-level SMC with BESS degradation details

4.1. Annual SMC loop modelling

4.1.1. Network optimization with BESS at contingencies

This section defines the network optimization stage at contingencies considering the BESS discharge as a corrective action after generation re-dispatch. This is simply an ACOPT with the main objective of minimizing total operation costs (TOC) from generation costs (TGC) and interruption costs (IC_{sys}) as modeled by (3).

Where the generation cost TGC_i for each generator $i \in Ng$ with power output P_i is represented as a quadratic function with coefficients a , b and c . The system interruption costs IC_{sys} are the sum of the interruption costs of each bus $j \in Nb$ at time t , which are a function of the interrupted energy assessment rate (IEAR) that represents the amount in \$ per each interrupted kWh with the load curtailment being the base energy not supplied without BESS utilization ($ENS_j^{base}(t)$) reduced by the discharged BESS power at time t , ($P_{j,BESSdis}(t)$). The minimization function (3) is subjected to the set of constraints calculated by (4) to (12).

$$\text{Min TOC}(t) = \text{Min}[TGC(t) + IC_{sys}(t)]$$

$$= \text{Min} \left[\sum_{i=1}^{i=N_g} \{a_i + b_i \times P_i(t) + c_i \times (P_i(t))^2\} + \sum_{t=t_0}^T \sum_{j=1}^{j=N_b} ENS_{j,base}(t) - P_{j,disch}(t) \times IEAR_j \right] \quad (3)$$

$$P_{i,min}^g \leq P_i^g \leq P_{i,max}^g \quad (4)$$

$$P_{ji} \leq P_{ji}^{max} \quad (5)$$

$$v_{min} \leq v_j \leq v_{max} \quad (6)$$

$$E_{j,BESS}(t) = E_{j,BESS}(t - \Delta t) + (\eta_{BESSch} \cdot P_{j,BESSch}(t) - P_{j,BESSdis}(t) / \eta_{BESSdis}) \cdot \Delta t \quad (7)$$

$$(P_j^D(t) - P_j^{curtailment}) + \eta_{BESSch} \cdot P_{j,BESSch}(t) + \sum_{ji} P_{ji}(t) = P_j^g(t) + \frac{P_{j,BESSdis}(t)}{\eta_{BESSdis}} \quad (8)$$

$$0 \leq P_{j,BESSch}(t) \leq P_{j,cap} \quad (9)$$

$$(ENS_j^{base}(t) - P_{j,BESSdis}(t) / \eta_{BESSdis}) \geq 0 \quad (10)$$

$$P_{j,BESSdis}(t) \times P_{j,BESSch}(t) = 0 \quad (11)$$

$$\sigma_{min} \times E_{j,cap} \leq DOD_j(t) \leq \sigma_{max} \times E_{j,cap} \quad (12)$$

Eqs. (4) to (6) determine the generation output limits, maximum line

flow limits, and bus voltage limits v_j . Eq. (7) calculates the available stored energy of BESS connected to bus j ($E_{j,BESS}(t)$) considering the state of charge at the previous time step and any charging ($P_{j,BESSch}$) or discharging ($P_{j,BESSdis}$) powers with their corresponding efficiencies $\eta_{j,BESSch}$ and $\eta_{j,BESSdis}$. Eq. (8) describes the power balance constraint at bus j at time t . Such that the left-hand side (L.H.S) of the equation shows the summation of the original demand at bus j reduced by any load curtailments, BESS charge and the summation of all line flows from bus j to bus i . This L.H.S is balanced with the right-hand side (R.H.S) of the equation which models the total power generation at node j from conventional generation and BESS discharge. Eq. (9) limits the charge power at any time step by the power rating of the BESS ($P_{j, cap}$) and similarly for the discharge. Eq. (10) ensures that the amount of BESS discharge won't exceed the energy not supplied at contingencies to avoid unwanted utilization of BESS. Eq. (11) guarantees that no charging and discharging could occur at the same time step. In (12), the minimum and maximum DOD limits, σ_{min} and σ_{max} , at time t are constrained as a percentage of the BESS energy capacity.

4.1.2. BESS degradation model

In this section the mapping of the degradation effect of the BESS either due to operation at nominal or elevated DOD levels is illustrated. This degradation effect is captured through a two-step process: (1) the estimation of the nominal DOD and (2) the equivalent capacity degradation.

(a) Identifying the Nominal BESS Depth of Discharge

A method for estimating the nominal operated DOD % for the BESS is illustrated below based on maximizing the total energy throughput of the battery.

This work presents a method for estimating the nominal DOD for the BESS based on maximising the total energy throughput of the battery. This uses the battery cycles against DOD curves (Fig. 1) to calculate using (13) the corresponding Ah throughput ($Ah_{i,throu}$) for each DOD_i . $Ah_{i,throu}$ is the multiplication of the DOD_i with the number of life cycles (N_i) corresponding to that DOD level and the ampere-hour capacity (Ah_{cap}).

$$Ah_{i,throu} = N_i \times DOD_i \times Ah_{cap} \quad (13)$$

Based on (13) the $Ah_{i,throu}$ for a discharge depth i is affected by the number of life cycles (N_i) at the DOD_i , and the ampere-hour capacity, Ah_{cap} , of the specified BESS. In Fig. 4 an example of the maximum $Ah_{i,throu}$ is shown for a DOD based on lithium ion battery data provided in [29], which can be used in the calculations as the nominal DOD (i.e., $DOD_{nominal}$); however, different BESS technologies will have a similar trend with different $DOD_{nominal}$.

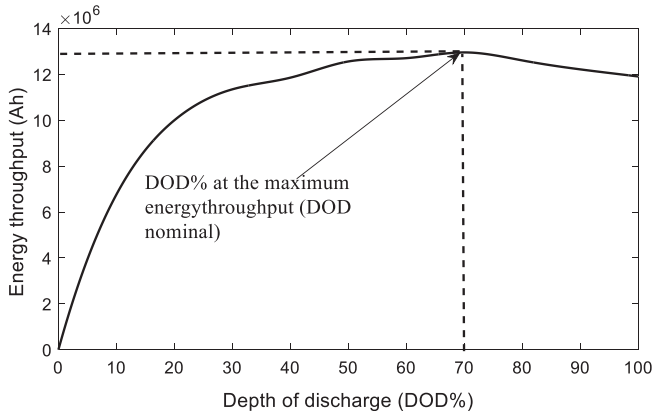


Fig. 4. BESS energy throughput against Depth of Discharge for Li-ion battery.

(b) BESS Equivalent Capacity Degradation Model

A different approach to calculate BESS degradation can be based on the Equivalent Capacity Degradation (ECD) model. The ECD model quantifies the different DOD impact on capacity degradation from the manufacturer capacity degradation data, with an example shown in Fig. 2, assuming also that EOL is reached when the battery capacity is reduced to 80% [28]. The ECD methodological steps are illustrated below.

- (1) The DOD of percentage i ($DOD_{j,i}$) for the k^{th} cycle of a BESS at bus j , is evaluated based on the change of stored energy at every Δt by (14) after the network optimization. In (14) tc_{start} and tc_{end} are the times that the discharge cycle starts and ends, respectively, as it might last for more than one Δt .

$$DOD_{j,i}(k_c) = \sum_{tc_{start}}^{tc_{end}} (E_{j,BESS}(t) - E_{j,BESS}(t + \Delta t)) / E_{j, cap} \times 100 \quad (14)$$

- (2) When a discharge cycle ends at a given $DOD_{j,i}$ an allocated counter $C_{DODj,i}$ is formed to record the number of discharge cycles occurred at this $DOD_{j,i}$. Hence several counters are implemented to record the frequency of different $DOD_{j,i}$ events during the BESS operation life cycle.
- (3) When any $C_{DODj,i}$ increases $C_{j,eq}$ is calculated by (15), which indicates counter of the equivalent number of cycles referred to the manufacturer discharge depth, DOD_{base} (Fig. 2). Hence, it captures the cumulative equivalent cycle degradation. $L_{DOD_{base}}$ indicates the number of life cycles the EOL is expected at DOD_{base} while L_i indicates the number of life cycles the EOL is expected at $DOD_{j,i}$. This cycle degradation is factorized in normal degradation ($C_{j,eq,norm}$) when $DOD_{j,i} \leq DOD_{j,nominal}$ and accelerated degradation ($C_{j,eq,acc}$) when $DOD_{j,i} > DOD_{j,nominal}$ by using (16).

$$C_{j,eq} = \sum_i (C_{DODj,i} \times L_i) / L_{DOD_{base}} \quad (15)$$

$$C_{j,eq} = \underbrace{\sum_{i \leq nominal} \frac{C_{DODj,i}}{L_{j,i}} \times L_{DOD_{base}}}_{C_{j,eq,norm}} + \underbrace{\sum_{i > nominal} \frac{C_{DODj,i}}{L_{j,i}} \times L_{DOD_{base}}}_{C_{j,eq,acc}} \quad (16)$$

- (4) At the end of each t and when there is an increase in any $C_{DODj,i}$ (i.e. indication for a new cycle), the equivalent capacity degradation, $Cap_{j,deg}$, based on the $C_{j,eq}$ is calculated from manufacturer curves (e.g., Fig. 2). Hence, $Cap_{j,deg}$ in (17) is also factorized to normal and accelerated to allow for capturing the effect of degradation due to normal DOD and due to higher DOD used during contingencies. Consequently, in (17) the Eq. (1) is used and replacing $N_{DOD_{base}}$ with $C_{j,eq,norm}$ to calculate $Cap_{j,deg,norm}$ and with $C_{j,eq,acc}$ to calculate $Cap_{j,deg,acc}$.

$$Cap_{j,deg} = f(C_{j,eq,norm}) + f(C_{j,eq,acc}) \quad (17)$$

$Cap_{j,deg}$ value is reduced from the original capacity of the battery to calculate the maximum charging point for the next charging cycle, within the network analysis.

A simple example is illustrated below to show the evaluation of $C_{j,eq}$ in Eq. (15). The manufacturer DOD for capacity degradation (DOD_{base}) is 100% for certain BESS technology as shown in Fig. 2. The current DOD_i for the k^{th} cycle is 80% with L_{80} of 1200 cycles. $C_{DOD,80}$ stores 7 cycles. The previously operated i^{th} DOD are 90% with $L_{90} = 960$ cycles and 75% with $L_{75} = 1540$ cycles where $C_{DOD,90}$ stores 5 cycles and $C_{DOD,75}$ stores 9 cycles. Based on Eqs. (15) and (16) the numerical value of $C_{j,eq,acc}$ is captured by (18) when DOD_i exceeds the nominal value which is 70% based on the example of Fig. 4.

$$C_{j,eq_acc} = \frac{C_{DOD,75}}{L_{75}} \times L_{100} + \frac{C_{DOD,80}}{L_{80}} \times L_{100} + \frac{C_{DOD,90}}{L_{90}} \times L_{100} \quad (18)$$

$$= \frac{9}{1540} \times 830 + \frac{7}{1200} \times 830 + \frac{5}{960} \times 830 = 14 \text{ cycles}$$

From (18) it can be seen that after a specific cycle, the equivalent accelerated cycles at 100% DOD are 14 cycles accounting for the current and the history of accelerated operation (i.e. $C_{j,eq_acc} = 14 \text{ cycles}$).

Now, the accelerated capacity degradation (Cap_{j,deg_acc}) is evaluated by substituting C_{j,eq_acc} with C_{eq} in (1). The numerical value of Cap_{j,deg_acc} is 0.16% when using values of $n = 2$, $a = 8.8 \times 10^{-7}$, $b = -0.018$ and $Z = 100$ for the parameters in (2) for a lithium-ion BESS data provided in [23]. This calculated value of Cap_{j,deg_acc} represents the reduction of the capacity considering the current and history of accelerated operation above the nominal DOD.

4.1.3. Annual SMC loop outputs

The annual calculation outputs are split into two groups the Network indices and BESS indices. The network performance indices used here are the expected energy not supplied (EENS), expected interruption costs (EIC) and expected total operating costs (ETOC) [2]. The BESS performance indices are created for every bus j on the network that has a BESS and describes the degradation risks. All indices are calculated at the end of each year and for number of years required to achieve convergence based on 5% in the EENS.

(a) Expected equivalent capacity degradation (EECD_j)

The $EECD_j$ captures the capacity degradation at both normal and accelerated $DOD_{j,i}$ operation by (19), in MWh/year. It can provide network planners with the expected BESS EOL and facilitate their long-term planning considering network uncertainties. It is divided into normal (i.e., $EENCND_j$) and accelerated (i.e., $EEACD_j$) capacity degradation by (20) and (21).

$$EECD_j = \sum_{y=1}^Y (Cap_{j,deg_norm}(y) + Cap_{j,deg_acc}(y)) / Y \quad (19)$$

$$EENCND_j = \sum_{y=1}^Y Cap_{j,deg_norm}(y) / Y \quad (20)$$

$$EEACD_j = \sum_{y=1}^Y Cap_{j,deg_acc}(y) / Y \quad (21)$$

(b) Expected equivalent life cycle accelerated degradation

The Expected equivalent life cycle accelerated degradation ($EELCAD_j$) captures the annual degradation due to elevated $DOD_{j,i}$ operation by (22), in cycles/year. It only considers the cycles BESS operated at $DOD_{j,i}$ above $DOD_{j,nominal}$. It is obtained by calculating the equivalent accelerated cycle degradation $ELCAD_{j,y}$, from (23), referred to the $DOD_{j,nominal}$ operation estimated in Fig. 4.

$$EELCAD_j = \sum_{y=1}^Y ELCAD_{j,y} / Y \quad (22)$$

$$ELCAD_{j,y} = \sum_{i>nominal} C_{DOD_{j,i}}(y) \times L_i / L_{DOD_{j,nominal}} \quad (23)$$

In (23) $C_{DOD_{j,i}}(y)$ is the $C_{DOD_{j,i}}$ value at the end of SMC year (y). Using the same concept in (23) and (24), the Expected equivalent life-cycle normal degradation (EELCND) could be captured at $i \leq nominal$.

(c) Expected BESS accelerated degradation costs (EBADC_j)

$EBADC_j$ translates the accelerated life-cycle degradation to cost, in M \$/year using (24), which allows the operators to quantify the economic risks of the accelerated degradation of BESS during contingencies.

$$EBADC_j = \left[\sum_{y=1}^Y C_{j,capital} \times (ELCAD_{j,y} / L_{DOD_{j,nominal}}) \right] / Y \quad (24)$$

In (24) $C_{j,capital}$ is the capital cost of BESS at bus j . Hence, $EBADC_j$ helps to quantify the costs of annual accumulated accelerated BESS degradation as a portion of its capital costs.

4.2. Life-Cycle SMC loop modelling

In the 2nd SMC life cycle loop, the entire accelerated degradation history of the BESS at each bus j is captured sequentially through a defined coefficient $\mathfrak{R}_{j,Y_L}$. $\mathfrak{R}_{j,Y_L}$ is determined by (25) considering the expected normal BESS degradation as well as the expected increased risk of BESS accelerated life cycle degradation at emergencies.

$$\mathfrak{R}_{j,Y_L} = \sum_{Y_L=1}^{Y_{Life}} EELCAD_{j,Y_L} + \sum_{Y_L=1}^{Y_{Life}} EELCND_{j,Y_L} \quad (25)$$

Y_{Life} in the above equation is the BESS estimated life in years.

5. Time varying thermal ratings

According to the IEEE standard 738 [37], TVTR is affected by the dynamic weather conditions from the wind speed (V_m), wind direction (K_{angle}) and ambient temperature (T_a) as well as the solar radiations. The IEEE thermal rating model is based on the thermal equilibrium of the OHL conductor. Such that the convection heat loss (q_c) and radiated heat loss (q_r) are equalized by the solar heat gain (q_s) and the heat gain by the Joule losses due to the current flow in the conductor (I) at a certain conductor temperature T_c . The thermal equilibrium is developed by (26) and then the current flow could be evaluated by (27). The calculations of q_c , q_r , q_s and R are clearly illustrated in the IEEE-738 standard.

$$q_c(t) + q_r(t) = q_s(t) + I(t)^2 x R(T_c(t)) \quad (26)$$

$$I(t) = \sqrt{\frac{q_c(T_c, T_a, K_{angle}, V_m, t) + q_r(T_c, T_a, t) - q_s(t)}{R(T_c, t)}} \quad (27)$$

From (27) the dynamic line ratings (in Amperes) are updated each hour based on the real time hourly weather data.

6. Case study design

The proposed methodology is applied on the 24 bus IEEE-RTS system with the provided chronological hourly load data per each bus and an increased demand level to 1.7 pu [38]. All the network cables are modeled with OHL properties. The simulation stopping criterion is set to 5% EENS covariance and 2500 years, whichever is achieved first, while SMC simulation considers only line failures. The optimization is implemented using the MIPS solver of MATPOWER within MATLAB [39]. It should be mentioned that in this analysis the BESS is assumed to be owned by the network operator with the BESS operating scenarios summarized in Table 1. It is also assumed that battery cells are aggregated to one large BESS at each bus with identical cell characteristics. An 1% of annual load growth is considered in the life cycle SMC loop.

6.1. BESS technology and costs

A lithium-ion BESS is considered in the study with the details shown in Table 2. The total BESS costs per bus are evaluated by (26) and the cost data in Table 2.

$$C_{capital}^j = P_{j,cap} \times [C_{j,PCS}] + E_{j,cap} \times [C_{j,ESM}] \quad (26)$$

Table 1

Details of study scenarios.

Scenarios	Scenario description
Base case	No BESS is installed in the network
Sc-1	BESS is installed in IEEE-RTS without considering capacity degradation but constrained DOD up to nominal value.
Sc-2	BESS is installed in IEEE-RTS without considering capacity degradation but un-constrained DOD at (i.e up to 100%).
Sc-3	BESS installation as per Sc-1 but considering normal DOD degradation (Normal Capacity degradation is considered).
Sc-4	BESS installation as per Sc-1 but considering accelerated DOD degradation.(Accelerated capacity degradation is considered).

Table 2

Lithium Ion BESS Design and Operating Specification.

Term	Lithium ion
Charge and discharge efficiency	$\eta_{j,BESSch} = 80\%$, $\eta_{j,BESSdisch} = 80\%$ [28]
Capacity degradation parameters for Eq.c (1)	$n = 2, a = 8.8 \times 10^{-7}, b = -0.018, c \& d \& e = 0, Z = 100$ [40]
Nominal DOD	70 % (Fig. 4)
Nominal life time	3000 cycles for 10 years [29]
Maximum DOD	100 %
σ values for (11)	$\sigma_{min} = 0, \sigma_{max} = 100$
Per unit cost parameters	500 \$/KW, 890 \$/kWh [41,42]

In (26) $C_{j,PCS}$ is the capital cost for the power conversion system and $C_{j,ESM}$ is the capital costs of the energy storage medium at bus j , while $P_{j,cap}$ and $E_{j,cap}$ are the power and energy capacities for the BESS facility at bus j .

6.2. BESS location and size

The prioritization of bus at which BESS installation is considered to contribute most to the network operation is performed based on the total network operation and interruption costs reduction (TOC_{RE}) and assuming the installation of a single BESS. The calculated TOC_{RE} values in Table 3 are produced considering a single BESS of 70 MWh and 50 MW installed at the specified bus. However, the use of other fixed sizes ends up to the same ranking trend but with different values of TOC_{RE} . Hence, B6 is the most optimum location to install a BESS in IEEE-RTS due to the highest TOC_{RE} produced (Table 3) while the B22 is the least significant bus to install a BESS, using this approach.

The total BESS size at each bus, i.e., power rating and energy capacity quantities, is calculated considering the 20% of the bus annual peak demand and 2% of the bus annual peak day energy consumption [36], although other values could be used.

The total BESS installed in the network is calculated by (24) and assuming a maximum available BESS investment cost of 230 M\$. Hence, the BESS for the IEEE-RTS network is shown in Table 3 with a total cost of 223 M\$ for the first three locations in the ranking list (B6, B8 and B14) which is covered by the assumed available BESS investment of 230 M\$. This 223 M\$ is divided into energy storage medium costs of 160 M\$ and power conversion system cost of 55 M\$ based on the cost data of Table 2. The power conversion system (due to its EOL) is set to be replaced after

Table 3

BESS Location Ranking Installation and Size.

Bus No	B6	B8	B14	B16	B4	B1	B15
TOC_{RE} (M\$/y)	47.8	40.4	36.7	31.3	29.6	28.2	25.4
$E_{j,cap}$ (MWh)	44.4	63.3	71.8	—	—	—	—
$P_{j,cap}$ (MWh)	27.8	45.32	51.32	—	—	—	—
Bus No	B7	B5	B19	B9	B18	B13	B20
TOC_{RE} (M\$/y)	22.6	19.4	15.8	11.9	9.8	9.1	8.3
						1.7	
$E_{j,cap}$ (MWh)	—	—	—	—	—	—	—
$P_{j,cap}$ (MWh)	—	—	—	—	—	—	—

20 years [43].

7. Modeling outputs and discussion

The overall values of the network and BESS performance indices are captured in this section.

7.1. Annualized network and BESS performance indices

Table 4 summarizes the network reliability and BESS degradation indices for the different studied scenarios. The best network reliability is recorded in Sc-1 when BESS is installed in the network. This is mainly due to the higher available BESS capacity in the system since no capacity degradation is considered (i.e., BESS indices in Table 4 are null). The calculated reduction of 31% in EENS, 46% in EIC and 21% for ETOC is achieved which is a significant improvement against the no BESS base case (BC) scenario. In Sc-2 a further improvement of the indices is indicated against Sc-1, since degradation of the BESS is neglected and the DOD is un-constrained (i.e., to 100%) which gives more contingency support and hence better reliability indices from Sc-1.

When considering the BESS degradation at normal DOD (Sc-3) the network reliability performance is reduced when compared to both Sc-1 and Sc-2, although it is still better than not having BESS in the network. This is the result of constraining the DOD of the battery to 70% as well as considering the capacity degradation after each cycle which reduces the amount of energy support from the BESS.

In Sc-4, the network reliability indices are reduced (i.e., reliability is improved) compared to Sc-3 such that EENS, EIC and ETOC are reduced by 11%, 30% and 29% respectively. This is because the BESS is allowed to be cycled at higher DOD percentages than the nominal DOD so more power could be discharged at contingencies but with the downside of accelerated BESS degradation.

All the battery performance degradation indices are null for the base case, Sc-1 and Sc-2 as no BESS in the base case and in Sc-1 and Sc-2 the battery life is assumed constant with no degradation consideration. On the other hand, for Sc-4 the EELCAD is recorded as 158 cycle/year. Moreover, the EBADC is recorded in Sc-4 as 8.92 M\$/yr.

Looking at the EENCD and the EEACD, it is found to be 3.17 and 1.21 MWh/year. This reflects the impact of the accelerated DOD operation of the BESS in extra 1.96 MWh of expected annual capacity reduction which results in life-time reduction. Network planners could use this index to estimate the expected BESS replacement time when the capacity falls by 20% of the original value. Hence, they could assess the expected early replacement risks versus the expected reliability benefits.

Based on the above results in Table 4, there is overestimation of the reliability benefits when no BESS degradation is considered (Sc-1 and Sc-2). This leads to underestimation of the degradation risks of the BESS and their expected replacement time as their life is assumed constant ending to an inaccurate network planning with BESS consideration.

7.2. The sensitivity of EIC and EBADC against DOD%

A sensitivity analysis is conducted for a range of DOD from (70% till

Table 4

Network Reliability and BESS Degradation Outputs.

	Indices	Scenarios				
		BC	Sc-1	Sc-2	Sc-3	Sc-4
Network	EENS (GWh/yr)	10.3	7.09	6.39	7.93	7.21
	EIC M\$	42.8	22.08	20.27	33.19	22.96
	ETOC (M\$/yr)	930	809.42	784.46	872.43	845.36
BESS	EELCAD (Cyc/yr)	0	0	0	0	158
	EELCND (Cyc/yr)	0	0	0	236	106
	EENCD (MWh/yr)	0	0	0	2.98	3.17
	EEACD (MWh/yr)	0	0	0	0	1.21
	EBADC (M\$/yr)	0	0	0	0	8.92

100%) such that the 70% represent the nominal DOD level. The analysis is conducted from 70% till 100% DOD in 5% steps and then the expected interruption costs (EIC), the expected battery accelerated degradation costs (EBADC) are captured for each DOD%. Fig. 5 plots the relation of EIC and EBADC against the DOD% range. The secondary vertical axis shows the degradation benefit in M\$ as the difference between EIC reduction and EBADC increase.

From the figure, it is obvious that the more accelerated DOD operation, the more reduction in EIC and increase in EBADC. Also, the degradation benefit in M\$ increases with accelerating the DOD operation reaching its maximum value at 90% DOD then starts to be reduced. This means that the accelerated degradation value is reduced after 90% DOD. This is as the operation of BESS above 90% DOD on all occasions might not be always required and hence more degradation occurs with no need (i.e., extra accelerated degradation costs with no benefits). Hence, it is very important to operate the BESS up to a maximum DOD limit to avoid unnecessary degradation.

7.3. BESS life cycle analysis with degradation impact on NPV

In this section, a planning analysis is conducted over the nominal life time of the BESS of 10 years (Table 2). Such that the cumulative annual life-cycle benefits from BESS degradation are assessed in terms of comparing the degradation risks, early replacement, with the reliability benefits.

The annual savings from BESS, $Saving_{yL}$, are captured by (27) as the annual expected reduction in the total network operation costs (TOC_{RE}) reduced by the annual EBADC if accelerated degradation due to elevated DOD operation is considered.

$$Saving_{j,yL} = TOC_{RE,j,yL} - EBADC_{j,yL} \quad (27)$$

Fig. 6 shows the annual savings from BESS operation at both Sc-3 and Sc-4. It is clear that the annual savings from BESS are higher in Sc-4 than Sc-3 because of the higher reduction in the interruption costs as a result of high elevated DOD operation. In addition, the payback period for the BESS capital costs is found to be 8 years in Sc-3 and 5 years in Sc-4. This defines the break point where the cumulative annual savings at both scenarios are equal to the BESS capex (223 M\$). The benefits of accelerated degradation are also shown through the hashed area A1 in Sc-4 representing the cumulative savings after the payback which is higher than the corresponding area A2 for Sc-3.

On the other hand, the BESS needs to be replaced at Sc-4 two years earlier compared to Sc-3 as a result of the accelerated degradation such that the accelerated capacity degradation let to reaching the 80% of the capacity 2 years earlier than Sc-3.

To clearly show if the accelerated degradation is worth or not, the net present value (NPV) is calculated for each scenario using the capex of 223 M\$ and an interest rate of 5%. In the NPV calculation, if the BESS is replaced before the end of the planning period the cost of replacement is split on the BESS nominal life and only the remaining years from replacement till the end of planning period are considered.

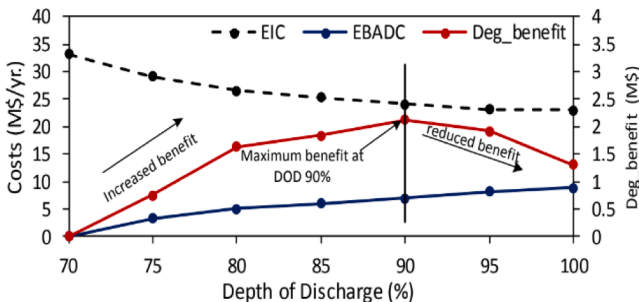


Fig. 5. Effect of DOD on the Expected Interruption costs and Expected BESS accelerated degradation costs.

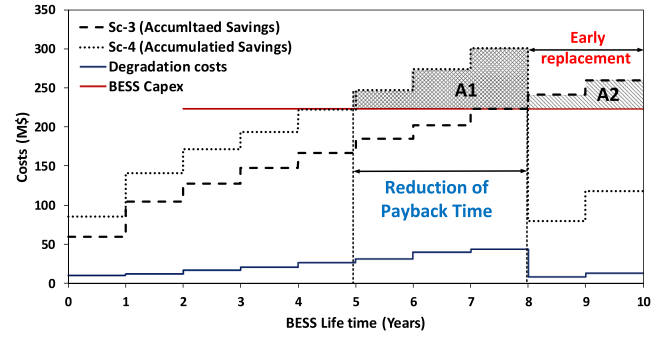


Fig. 6. Life cycle loop showing the risks/benefits from BESS degradation.

The NPV for Sc-3 is (-11.06 M\$) and for Sc-4 is (31.53 M\$). The positive NPV means profits which shows that accelerating the degradation of the BESS to have more techno-economic reliability benefits have a great impact on the profits of the network although the BESS is replaced earlier. However; to make Sc-3 of normal degradation profitable in the planning analysis a cheaper BESS should be used (assuming all other variables unaffected).

If the cost of the energy storage medium is below 830 \$/KWh this will result in a positive NPV for Sc-3. Hence at 830 \$/KWh the capex is 212 M\$ which results in zero NPV at Sc-3 and 42.53 M\$ NPV profits in Sc-4. It should be mentioned that for a Li-ion BESS technology a cost of 830 \$/KWh is a reasonable and already feasible as mentioned in [43] showing the range of \$/KWh price for different BESS technologies. Fig. 7 shows the variation of NPV with the storage medium cost in \$/KWh for Sc-3 and Sc-4.

The above analysis clearly reflects the importance of inclusion of accelerated BESS degradation in the economic planning analysis and record the effect on NPV profits.

7.4. Network long-term planning analysis

In this section, the results of long-term planning over different network planning horizons are illustrated showing the number of BESS replacements over the planning horizon.

Table 5 shows the number of BESS replacements (β) and the NPV for different network planning horizons from 10 up to 30 Years for both Sc-3 (normal degradation) and Sc-4 (accelerated degradation). From Table 5 it is clear that long-term network planning considering Sc-4 is much better than Sc-3 in terms of NPV context although with more BESS replacements.

The zero crossing for positive NPV is 830 \$/kWh for the 10 years planning, 600 \$/kWh for the 20 years planning, and 390 \$/kWh for the 30 years planning period. It should be noted that price of \$/kWh for the BESS ranges from 550 \$/kWh up to 1200 \$/kWh as per [43]. Hence, the cheaper the BESS technology the more profitable long-term planning models with higher potential for grid scale BESS deployment for

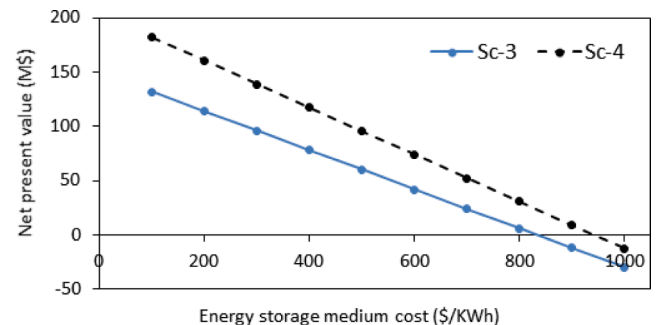


Fig. 7. NPV at different \$/kWh energy storage cost for Sc-3 an Sc-4.

Table 5NPV and BESS replacements (β) at different planning horizons.

Planning Horizon	BESS replacements (β)		NPV (M\$)	
	SC-3	SC-4	SC-3	SC-4
10	0	1	-11.06	31.53
20	1	2	-40.94	-7.12
				25.4
30	2	3	-194.31	-127.914

NOs.

7.5. Combined DTR with BESS impact on network reliability

This section examines the additional flexibility to the network when DTR is used along with BESS. Hence, the reliability indices are captured with BESS only, with DTR only, and with both BESS and DTR utilised at contingencies. The selected OHLs for TVTR implementation are 11,12,13,23 and 28, which represent the highly loaded lines within the network with a total length of 248 km [38].

It can be seen from Table 6 that the best reliability enhancements recorded when both BESS and DTR are utilised at contingencies.

The below steps are implemented to reflect more how DTR could be beneficial in reducing the size of BESS to preserve a desired network reliability level set by the NOs.

- A single aggregated BESS size is assumed with accelerated degradation aggregating all the BESS in the network with total energy capacity of 180 MWh and power rating of 124 MW with considering static thermal ratings (STR) in the ACOFF.
- The EENS is calculated as 7.21 GWh per year which is assumed as 1p.u (i.e. desirable value set by NO).
- The network analysis is repeated, and the EENS is evaluated considering DTR and BESS size reduced by a step of 2 MWh and 1 MW. The target is to preserve the same EENS calculated with SLR.
- The difference between the evaluated EENS and the required one is evaluated. If the difference is negative (i.e. EENS is lower than the required value) then the ESS size step is increased by 1 MWh and 0.5 MW. And if the difference is positive then the step is decreased by 1 MWh and 0.5 MW.
- The previous step is repeated till the EENS difference between two successive steps reaches 10 MWh as stopping criteria.

According to the findings, the use of DTR resulted in a decrease in the size of the BESS, with the capacity reduced to 155 MWh (from 180) and the power rating dropping to 114 MW (from 124) to maintain the same EENS as the one with solely using STR. This indicates a reduction of 25 MWh in energy capacity and 10 MW in power rating.

This reduction in the BESS size is equated to 27.45 M\$ less BESS capital costs investments as per the BESS cost parameters in Table 2. Assuming 17000\$/km for DTR implementation as per [42], then the total DTR capital costs is 4.21 M\$. Hence, DTR investment of 1.8% of BESS costs reduces 12.2% of the BESS capital costs to preserve the same NO desirable reliability level as per Fig. 8.

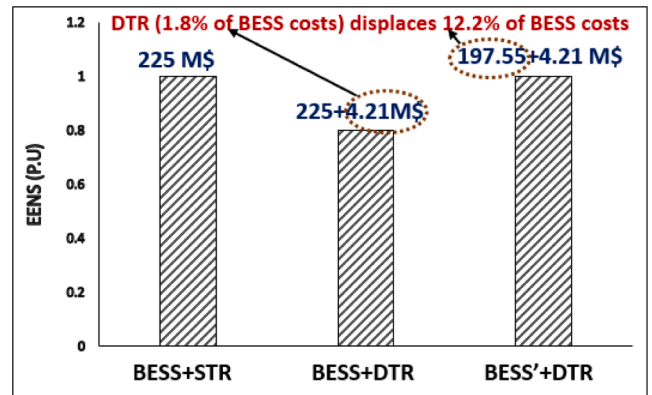
8. Conclusions

This paper presents a new methodology for assessing the risks and benefits of BESS accelerated degradation within the reliability framework. It maps the BESS degradation history and estimated lifetime risks, within a sequential multi-year network analysis over its useful life. Moreover, it provides a techno-economic planning analysis that allows for the assessing potential benefits and risks from BESS degradation on network reliability and NPV profits. To achieve this assessment three new indices are developed to quantify the BESS degradation risks; the expected equivalent capacity degradation (EECD), the expected

Table 6

Network Reliability Outputs.

Scenarios	EENS GWh/year	EIC (M\$/year)	ETOC (M\$/year)
BC	10.3	42.8	930
BESS	7.93	33.19	872.43
DTR	8.32	36.42	897.76
BESS + DTR	6.58	28.76	738.18

**Fig. 8.** DTR impact on reducing BESS size.

equivalent lifecycle accelerated degradation (EELCAD) and the expected equivalent degradation costs (EBADC).

The results show the benefits from BESS accelerated degradation due to reduced expected energy not supplied and expected interruption costs by 42% and 36% respectively, along with the positive impact of accelerated degradation on the NPV profits, even with the accounted increased risks of early replacements.

This method, and its measurable outcomes, can support the network planners' decisions on long-term network planning and the use of BESS for mitigating the effects of power system uncertainties.

Further work related to this research is integrating the risks of OHL ageing within the model and the value of BESS to reduce such risks and their effects from OHL maintenance interruptions. Hence, a full economic model could be developed gathering the OHL ageing costs, BESS degradation costs with the reliability costs. So, network operators could have guidelines on the value of temporarily stressing the lines and BESS assets for optimizing network reliability. This, however, will also require further improvements in their assets management and maintenance practices.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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